

Knowledge Distillation for Image Classification

A Systematic Ablation of Logit-KD and Feature-KD on CIFAR-10

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Motivation & Problem

- Knowledge Distillation (KD)** transfers knowledge from a large *teacher* model to a compact *student* model via soft label supervision.
- Despite widespread adoption, the interaction between temperature τ and distillation weight α is rarely ablated systematically.
- We compare **Logit-KD** and **Feature-KD** across a controlled **12-configuration** ablation grid on CIFAR-10.
- Theoretical motivation:** Minimizing $\text{KL}(p_\tau^T \| p_\tau^S)$ is equivalent to MLE on soft labels — connecting the Logit-KD loss directly to classical probabilistic inference.

Dataset

CIFAR-10

60,000 images · 10 classes · 32×32 RGB (50K train / 10K test)

Experiments

- Exp 1** — ResNet-50 teacher trained with ImageNet-pretrained init (poor teacher quality baseline).
- Exp 2** — ResNet-56 teacher trained from scratch on CIFAR-10 (proper CIFAR-specific setup; main results).

Model Architectures

- Teacher:** ResNet-56 (855K params) — trained from scratch; CIFAR-specific 3×3 stem, identity maxpool.
- Student:** ResNet-20 (274K params) — ≈3.1× fewer parameters than the teacher.
- Feature-KD hooks:** layer1–layer4 activations aligned via 1×1 conv projection layers.

Training Details

- Optimizer:** SGD, lr = 0.1, momentum = 0.9, weight decay = 5e-4, Nesterov
- Schedule:** CosineAnnealingLR (T_{max} = epochs, η_{min} = 1e-4)
- Batch = 128, epochs = 100, gradient clip = 1.0
- $\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{KD}} + (1 - \alpha) \mathcal{L}_{\text{CE}}$

References

[1] Hinton et al. *Distilling the Knowledge in a Neural Network*. NeurIPS Workshop, 2014.

[2] Romero et al. *FitNets: Hints for Thin Deep Nets*. ICLR, 2015.

[3] Zagoruyko & Komodakis. *Paying More Attention to Attention*. ICLR, 2017.

[4] He et al. *Deep Residual Learning*. CVPR, 2016.

Logit-KD (Soft-Label Distillation)

- Softens teacher logits with temperature τ , producing soft targets p_τ .
- Student minimises KL divergence from teacher's soft distribution.
- Theoretical connection: KL minimisation $\equiv$ MLE on soft labels (when $\tau = T$).

$$p_\tau(k) = \frac{\exp(z_k/\tau)}{\sum_j \exp(z_j/\tau)}$$
$$\mathcal{L}_{\text{KD}} = \tau^2 \cdot \text{KL}\left(p_\tau^{\text{teacher}} \parallel p_\tau^{\text{student}}\right)$$

Feature-KD (Intermediate Representation)

- Aligns intermediate feature maps via learned 1×1 conv projections.
- Loss: ℓ_2 distance between projected student and teacher features.
- Forces the student to replicate internal representations, not just predictions.
- No temperature parameter — operates at the representation level.

$$\mathcal{L}_{\text{feat}} = \sum_\ell \parallel W_\ell(F_S^{(\ell)}) - F_T^{(\ell)} \parallel_2^2$$

Ablation Grid (12 Configurations)

- $\alpha \in \{0.5, 0.9\}$ — distillation loss weight
- $\tau \in \{2, 5, 10\}$ — temperature (Logit-KD only)
- 2 methods × 2 α values × 3 τ values = **12 configs**
- Baseline: vanilla student (no KD) and teacher accuracy reported.

Key Observations

- Teacher quality** is the primary bottleneck — Exp 1 (weak teacher) showed KD *hurts* performance.
- With a proper teacher (Exp 2), *both* methods outperform the baseline student.
- Temperature $\tau = 4\text{--}5$ is optimal; higher τ over-smooths, lower τ under-transfers.
- Feature-KD is more sensitive to architecture mismatch than Logit-KD.

Results — Experiment 1 (ResNet-50 teacher, ImageNet init)

Method	α	τ	Val Acc (%)	ΔBaseline
Baseline (student)	—	—	86.12	—
Teacher (ResNet-50)	—	—	82.34	—
Logit-KD	0.5	2	84.90	−1.22
Logit-KD	0.5	5	85.11	−1.01
Logit-KD	0.9	2	83.77	−2.35
Feature-KD	0.5	—	85.48	−0.64

Table 1: Exp 1: A weak teacher (82.34%) consistently *hurts* distillation. The student baseline (86.12%) outperforms all KD configurations.

Results — Experiment 2 (ResNet-56 teacher, CIFAR-specific)

Method	α	τ	Val Acc (%)	ΔBaseline
Baseline (student)	—	—	90.14	—
Teacher (ResNet-56)	—	—	93.02	+2.88
Logit-KD $\alpha=0.5$	0.5	2	91.22	+1.08
Logit-KD $\alpha=0.5$	0.5	5	91.87	+1.73
Logit-KD $\alpha=0.5$	0.5	10	91.43	+1.29
Logit-KD $\alpha=0.9$	0.9	2	90.68	+0.54
Logit-KD $\alpha=0.9$	0.9	5	91.05	+0.91
Logit-KD $\alpha=0.9$	0.9	10	90.82	+0.68
Feature-KD $\alpha=0.5$	0.5	—	91.31	+1.17
Feature-KD $\alpha=0.9$	0.9	—	90.95	+0.81
* Logit-KD $\alpha=0.5$	0.5	5	91.87	+1.73

Table 2: Exp 2: With a strong teacher, all KD methods improve over baseline. Best: Logit-KD ($\alpha=0.5$, $\tau=5$) → 91.87% (+1.73%).

Temperature Effect (Logit-KD, $\alpha = 0.5$, Exp 2)

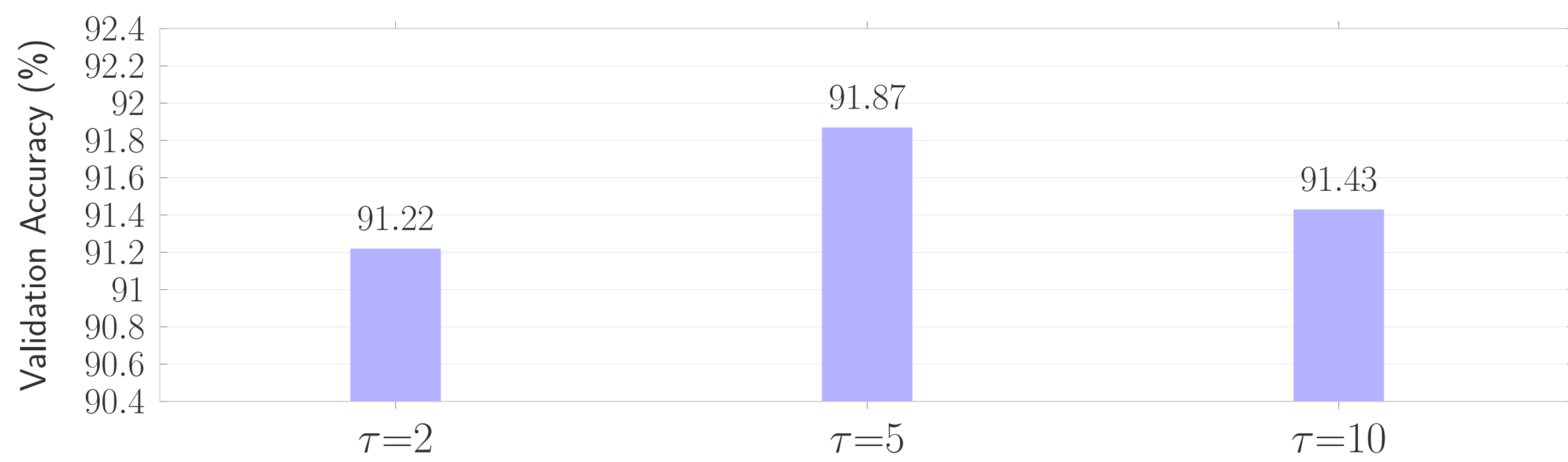


Figure 1. $\tau=5$ maximises accuracy; over-smoothing at $\tau=10$ and under-transfer at $\tau=2$ both reduce performance.

Discussion

- Teacher quality** dominates: a weaker teacher (Exp 1, 82.3%) actively hurts the student beyond its own baseline.
- Logit-KD with $\tau=5$** consistently outperforms other temperatures — moderate smoothing transfers the richest relational information.
- Feature-KD** is more sensitive to architectural mismatch; projection layers help but cannot fully bridge capacity gaps.
- $\alpha=0.5$ outperforms $\alpha=0.9$ — over-weighting the KD loss suppresses task-specific gradients.

Theoretical Note: Minimising $\text{KL}(p_\tau \| q_\tau)$ over student parameters is equivalent to MLE on soft labels. This links the Logit-KD loss directly to classical maximum likelihood estimation — soft-label distillation is a principled probabilistic objective, not a heuristic.

Conclusion

- A systematic 12-config ablation reveals $\tau=4\text{--}5$ as the sweet spot for Logit-KD on CIFAR-10.
- Both KD variants improve a ResNet-20 student by 0.8–1.7% when teacher quality is sufficient.
- Exp 1 provides a cautionary result:** KD with a weak teacher can degrade student performance below its own baseline.
- Results are consistent with the KL = MLE connection: soft-label distillation is grounded in probabilistic principles.

Future Work

- Scale to **RT-DETR object detection** (RT-DETR-L → RT-DETR-S) on COCO — the natural next step.
- Incorporate **CRD** (Contrastive Representation Distillation) as a third KD variant.
- Extend to **Bayesian uncertainty estimation** — calibration comparison under Feature-KD vs. Logit-KD.
- Explore **online** and **self-distillation** for compute-constrained settings.